Design and Implementation of VANITRA: A Wearable Assistive Device for Bidirectional Deaf–Mute Communication

*Note: This work was carried out exclusively for academic purposes and is submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering.

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***Abstract*—Millions of people who are deaf and mute face significant challenges in everyday communication, limiting their social participation and opportunities. This paper introduces *Vanitra*, a wearable device designed to facilitate two-way communication between deaf-mute users and hearing individuals. The system integrates multiple sensors—including flex sensors, hall effect sensors, magnets, and an inertial measurement unit (MPU6050)—with an ESP32-S3 microcontroller. Using sensor fusion and an embedded artificial intelligence model, sign language gestures are accurately recognized and converted into speech. Conversely, spoken words are transformed into tactile Morse code vibrations for the user. The offline and portable nature of Vanitra enables seamless communication without dependency on external networks. Initial prototype evaluations show the system's capability to provide reliable gesture recognition and effective bidirectional communication, promising enhanced accessibility for deaf-mute communities.**

***Keywords*—Wearable Assistive Device, Sensor Fusion, Sign Language Recognition, Artificial Intelligence, ESP32-S3, Haptic Feedback, Bidirectional Communication**

I. Introduction

Communication is a fundamental human need, yet millions of people worldwide face barriers due to hearing and speech impairments.[1][2] Deaf-mute individuals often struggle to interact with hearing people, limiting their social inclusion, education, and employment opportunities.[3] Traditional solutions, such as sign language interpreters or smartphone apps, have limitations like high cost, lack of portability, or dependence on internet connectivity.

To address these challenges, this paper presents *Vanitra*, a wearable bidirectional communication system that enables seamless interaction between deaf-mute users and hearing individuals. Vanitra uses sensor-equipped gloves with flex sensors, hall effect sensors, magnets, and an inertial measurement unit to recognize sign language gestures. An embedded artificial intelligence model running on an ESP32-S3 microcontroller converts these gestures into speech. Conversely, spoken words are transformed into Morse code vibrations through a haptic motor, allowing deaf-mute users to receive messages tactilely.

Vanitra operates fully offline, making it suitable for everyday use without reliance on network availability. This system aims to empower deaf-mute individuals with a portable, affordable, and effective communication tool.

II. Theroretical Background

A. Problem Statement

Deaf and mute individuals face substantial obstacles in communication, impacting their daily interactions, education, and employment opportunities. Many rely on sign language, but communication with hearing individuals often requires interpreters or technology that may be either expensive or unavailable.[4] Existing solutions largely focus on one-directional translation or require internet connectivity, limiting accessibility and independence. This communication gap contributes to social isolation, reduced workforce participation, and an overall decline in quality of life for users.[5] There is a clear need for a practical, offline, wearable system that supports bidirectional communication, empowering deaf-mute people to interact more seamlessly with the hearing community.

B. Motivation: The need for accessible, affordable bidirectional systems

Existing solutions, such as sign language interpreters, hearing aids, and smartphone applications are often expensive, inaccessible, or reliant on internet connectivity. Many deaf-mute

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individuals continue to face daily struggles in expressing themselves and understanding others, especially in environments where interpreters are not available or digital tools are impractical.[6] Limitations in current technology include lack of portability, inadequate bidirectional communication, and restricted usability in offline settings. Addressing these problems requires a practical approach that combines affordability, autonomy, and adaptability to diverse social and professional contexts.[5]

C. *WHO Statistics, Employment Data, and Market Gaps*

- Over 1.5 billion people globally experience various degrees of hearing loss.[1]
- Approximately 430 million suffer from disabling hearing loss that severely affects communication, including 34 million children.[1][2]
- By the year 2050, the number of people with disabling hearing loss is projected to exceed 700 million globally.[1]
- Data indicates only about 17% of individuals who could benefit from hearing aids currently use them, largely due to cost, access, and awareness barriers.[4][6]
- The economic impact of untreated hearing loss, accounting for healthcare costs, lost productivity, and societal consequences, is estimated at nearly US$980 billion annually.[6]
- In India, approximately 63 million people have hearing impairments; of those aged 15–59, only about 26% have stable employment, reflecting significant socioeconomic challenges within this population.[3][5]

D. *Brief About Vanitra and Its Innovation*

Vanitra is introduced as a wearable, bidirectional communication system designed to address these gaps.[5] The system features sensor-embedded gloves, equipped with flex sensors, hall effect sensors, magnets, and an MPU6050 inertial measurement unit controlled by an ESP32-S3 microcontroller. Vanitra employs sensor fusion and artificial intelligence to translate sign language gestures into speech output for hearing individuals. Conversely, spoken words are processed and transformed into tactile Morse code feedback via haptic motors, allowing deaf-mute users to receive messages through `vibration. The fully offline design ensures reliable, portable communication without dependence on external networks, making Vanitra an accessible and empowering solution for users in varied settings.

III. Literature Review

A. *Sensor-Based Sign Language Recognition Systems*

Over the past decade, research on sensor-equipped gloves for sign language recognition has progressed significantly.[7] Early systems, like those by Ahmed et al., used flex sensors with Arduino to detect finger movements and trigger simple audio responses but were limited to one-way communication and small vocabularies.[8] Lee & Lee enhanced this by combining flex, pressure sensors, and an inertial unit for ASL character recognition, supported by mobile apps but relying on connectivity.[9] Ghotkar et al. focused on Indian Sign Language using camera-based visual tracking, which reduced portability due to environmental constraints.[10] Vigneshwaran et al. developed flex sensor modules with Hidden Markov Models for English alphabets but mainly addressed static gestures, lacking support for dynamic signs needed for natural interaction.[11]

B. *Bidirectional Communication and Haptic Feedback*

Existing glove-based systems typically emphasize one-way translation from sign language to text or speech, neglecting the need for bidirectional interaction.[7] Haptic feedback approaches—such as Morse code vibration—have emerged as potential solutions for conveying speech information to deaf users. Despite this, few commercial or academic systems integrate both gesture interpretation and haptic feedback for hearing input, and fewer still perform all processing offline or autonomously.

C. *AI and Embedded Platforms*

Recent developments in embedded artificial intelligence and TinyML enable gesture and speech recognition on resource-constrained devices.[12] Systems utilizing ESP32 or similar microcontrollers can run machine learning models for real-time interpretation without requiring cloud connectivity. While promising, these solutions are still emerging in literature, and challenges remain in terms of model accuracy, dataset diversity, multi-language support, and battery life.[13]

D. *Feature Comparison Table*

Table 1. Comparison of Existing Solutions

System	Ahmed et al.	Lee & Lee	Ghotkar et al.	Vigneshwaran et al.	Vanitra (Proposed)
Sensors Used	Flex sensors (glove)	Flex, pressure, IMU	Vision + sensors	Flex sensors	Flex, hall, IMU, magnets
Bidirectional	No	No	No	No	Yes
Offline Use	Partial	No	Partial	Yes	Yes
Haptic Feedback	No	No	No	No	Yes
Main Limitation	Limited vocabulary	Needs app, not portable	Static alphabet only	Static alphabet only	Prototype stage

E. *Identified Gaps in Existing Solutions*

- Bidirectional Communication: Most systems only convert sign language to speech/text; few enable hearing speech to tactile feedback for deaf users.[8][14]
- Offline Functionality: Dependence on smartphones, PCs, or the internet hinders use in low-resource settings.[]
- Haptic Feedback: Rarely integrated, despite its potential for receiving information non-visually.[11]
- Dynamic Sign Recognition: Limited support for natural sign language, which often includes dynamic gestures and movement.[10]
- Portability and Accessibility: Many systems are bulky, require external cameras, or lack integrated AI capabilities.

IV. System Design and Architecture

A. *Glove Unit: Gesture to Speech*

The wearable glove enables deaf-mute users to convert sign language gestures into spoken words. Its components and functions include:

- Flex Sensors: Integrated along the fingers to detect the degree of bending, essential for gesture formation.
- Hall Effect Sensors and Magnets: Hall sensors on the thumb and palm, with magnets on the fingertips, accurately detect finger contacts and specific sign configurations.
- IMU (MPU6050): Mounted on the back of the glove or wrist, the inertial measurement unit tracks comprehensive hand movement and orientation for dynamic gesture interpretation.
- ESP32-S3 Microcontroller: Collects sensor data, calibrates and analyses it using embedded algorithms or AI models, classifies recognized gestures, and prepares output.
- Speaker: Receives processed data from the ESP32-S3 and articulates the detected word or phrase, enabling effective communication with hearing individuals.

B. *Speech Unit: Speech to Haptic Feedback*

This unit facilitates reception for deaf users by transforming spoken words into tactile signals:

- Microphone: Captures speech from a hearing person.
- ESP32-S3 Microcontroller: Processes incoming audio, converts recognized speech into Morse code (or other tactile language), and manages haptic feedback logic.
- Haptic Motor (Vibromotor): Outputs Morse code vibrations, allowing the wearer to receive and interpret messages via tactile sensations.

C. *System Overview*

a) CAD Hardware Layout:



Fig. 1. Glove Unit

Figure 1: This glove CAD shows a wearable made from flexible, comfortable textile such as neoprene or spandex. It features reinforced zones for securely mounting sensors (flex, hall effect, magnets) and the electronics module, while maintaining dexterity and a snug fit. The thicker wrist and palm sections help house the microcontroller and wiring, ensuring both durability and user comfort in daily wear.

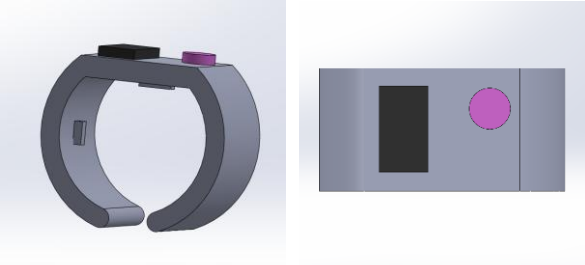


Fig. 2. Glove Unit

Figure 2: This CAD shows the speech module worn as a wristband. It is made from a lightweight, durable material for comfort and daily use. The pink part on top is the microphone, which picks up spoken words, while the black block is the ESP32-S3 microcontroller. On the inner side of the band, haptic motors are positioned to deliver vibration feedback directly to the user's skin. The band's design ensures all components are securely integrated while maintaining a discreet and ergonomic form.

D. *Labled Diagram*

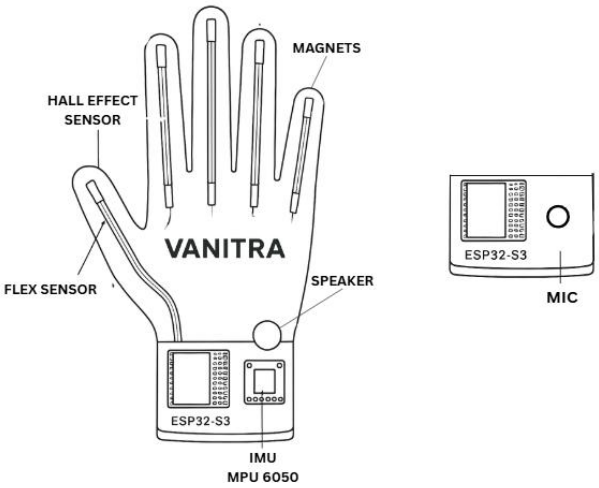


Fig. 3. Components Used and position

E. *Data Flow Description*

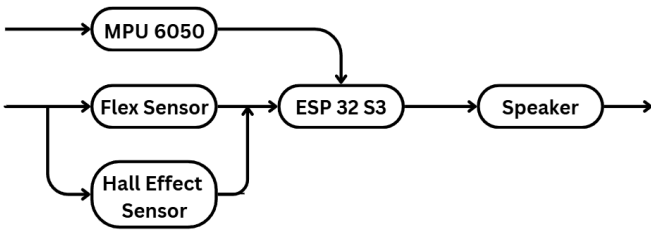


Fig. 4. Glove unit flow diagram

Glove Unit: User performs gesture → flex sensors, hall sensors, magnets, and IMU collect data → sensor fusion and AI classification on ESP32-S3 → decoded word sent to speaker.

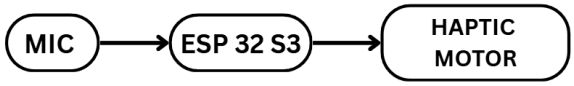


Fig. 5. Speech unit flow diagram

Speech Unit: Microphone captures speech → ESP32-S3 performs offline speech-to-text and translates it into Morse code → haptic motor vibrates to convey the message.

F. *Relevent Formula and Calculation*

a) Flex Sensor Normalization

Flex sensors give raw values based on finger bending. These raw readings may vary from sensor to sensor and user to user. Normalization helps scale all readings to a standard range (usually 0 to 1) so they can be processed by gesture recognition logic or AI.

$$x^{norm} = \frac{x_{measured} - x_{min}}{x_{max} - x_{min}}$$

$x^{norm} = The\ normalised\ Value\ between\ 0\ and\ 1$

$x_{measured} = Raw\ current\ Value$

$x_{min} = value\ when\ the\ finger\ is\ bend\ minimum$

$x_{max} = value\ when\ the\ finger\ is\ bend\ maximum.$

b) Hall Effect Sensor Detection

Hall effect sensors detect the presence of a magnet, which is used to sense if fingers are touching or close together (for certain signs).

$If\ V_{hall} > V_{threshold}, then\ contact\ is\ detected$

$V_{hall} = Voltage\ (or\ reading) from\ the\ hall\ effect\ sensor.$

$V_{threshold} = A\ pre-set\ threshold\ indicating\ the\ sensor\ has\ detected\ a\ magnetic\ field\ strong\ enough\ to\ signify\ contact.$

c) Hall Effect Sensor Detection

Convert text to Morse code, then encode it as vibrations using timing

(e.g., dot = short buzz, dash = long buzz).

V. METHODOLOGY

A. Sensor Calibration Process

Before data collection and gesture recognition, each sensor on the glove must be calibrated:

- Flex Sensor Calibration: Each flex sensor is calibrated by recording its output in two positions: finger fully straight (x_{min}) and fully bent (x_{max}). These values are used to normalize sensor readings, ensuring reliable gesture interpretation regardless of individual differences.
- Hall Effect Sensor Calibration: Hall effect sensors are tested with the corresponding magnet touching and not touching. The voltage threshold is established to reliably detect finger contact.
- IMU Calibration: The MPU6050 is calibrated for zero offset (stationary hand) and range, supporting consistent measurement of orientation and movement.

B. AI/ML Approach for Gesture Classification

- Sensor readings are preprocessed (normalized and sometimes filtered).
- A dataset of calibrated gestures is collected, with each gesture labeled according to its meaning.
- A supervised machine learning model (such as k-Nearest Neighbor, Random Forest, or a neural network) is trained to classify gestures based on sensor inputs.
- The trained model is embedded on the ESP32-S3 for real-time inference, outputting predicted phonemes or words as the user performs gestures.

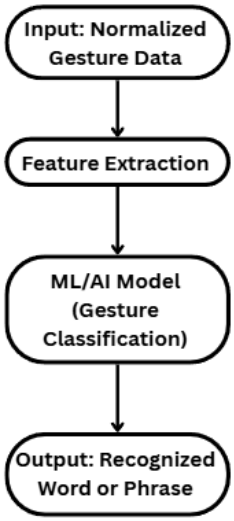


Fig. 6. Gesture to Phrase ML flowchart

C. Data Collection and Feature Extraction

- Data Collection: Multiple sets of gesture samples are recorded from different users to ensure robustness and generalization.
- Feature Extraction: Features may include normalized flex sensor values, hall sensor contact status, IMU angles and motion vectors. Statistical features (mean, variance, peak value) can be extracted as needed for improved accuracy and dynamic sign detection.

D. Mapping Gestures to Phonemes/Words

- Each gesture recognized by the AI model is mapped to a specific phoneme, word, or phrase in the system dictionary.
- The mapping ensures that the decoded hand movement corresponds to its intended spoken output. The ESP32-S3 interprets the AI model's output and triggers the speaker or buzzer accordingly.

E. Speech-to-Morse-code Conversion for Haptic Feedback

- In the second unit, speech input from the microphone is converted to text using a lightweight speech-to-text algorithm.
- The recognized text is encoded into Morse code using standard conventions (dot = short pulse, dash = long pulse).
- The encoded sequence controls the haptic motor's vibrations, transmitting the message to the user via touch.

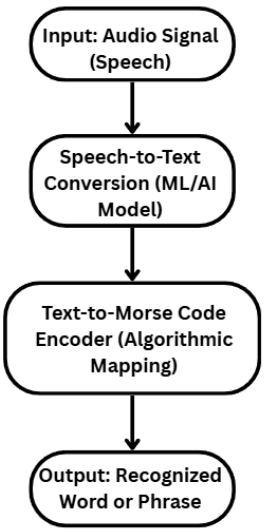


Fig. 7. Speech to Morse code ML flowchart

F. *Electrical Schematic Description*

The electrical schematic integrates all major system components:

- Flex and hall sensors are connected to ADC and digital inputs on the ESP32-S3.
- IMU (MPU6050) communicates via I2C protocol.
- Speaker/buzzer is connected to output GPIO with a suitable transistor or driver circuit.
- Microphone and haptic motor in the second unit are linked through input and output channels, managed by the ESP32-S3.
- Power supply, voltage regulation, and PCB layout ensure stable operation and user safety.

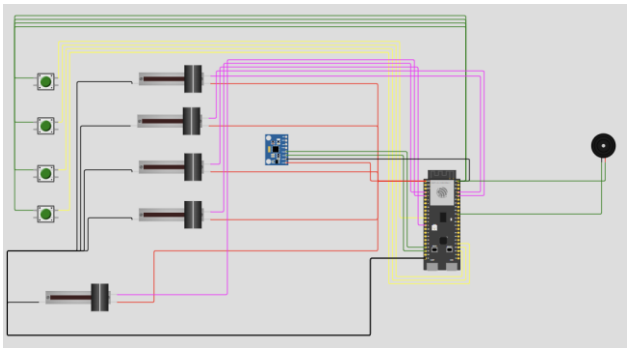


Fig. 8. General Electrical Schematic of glove unit.

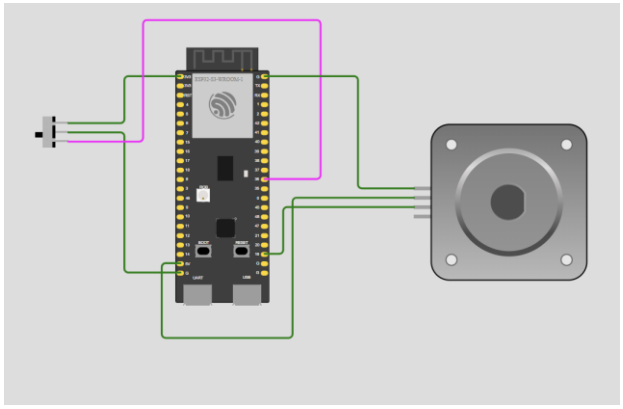


Fig. 9. General Electrical Schematic of Sppech unit.

VI. Results And Discussion

A. *Prototype Testing Setup*

The initial prototype was evaluated in both virtual and hardware environments:

- Wokwi Simulator: Used for preliminary validation of sensor signal processing, logic implementation, and system workflow. This enabled rapid iteration and debugging of the ESP32-S3-based firmware.
- Hardware Bench: The fully-assembled system—including the glove unit and the speech-to-haptic module—was tested on the bench using actual sensors, microcontroller, and output devices to assess real-world performance.

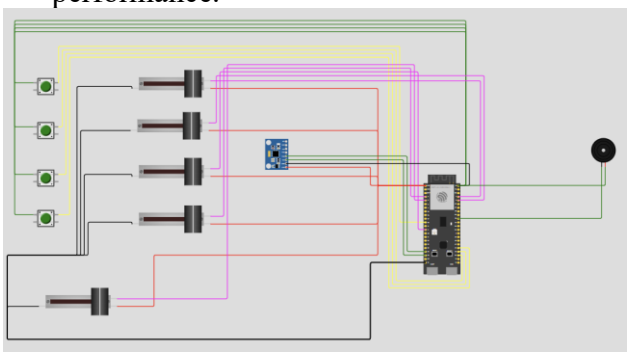


Fig. 10. WokWi Simulation of Glove Unit.

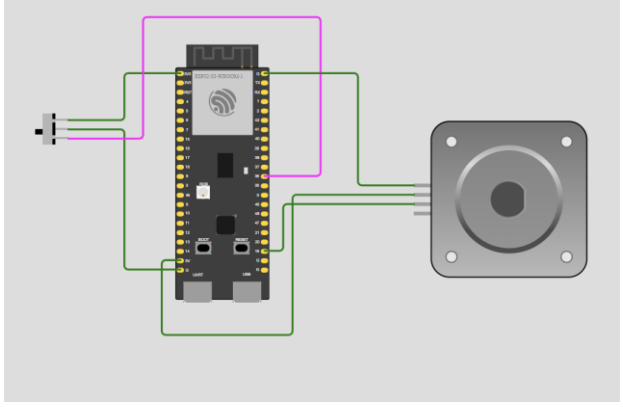


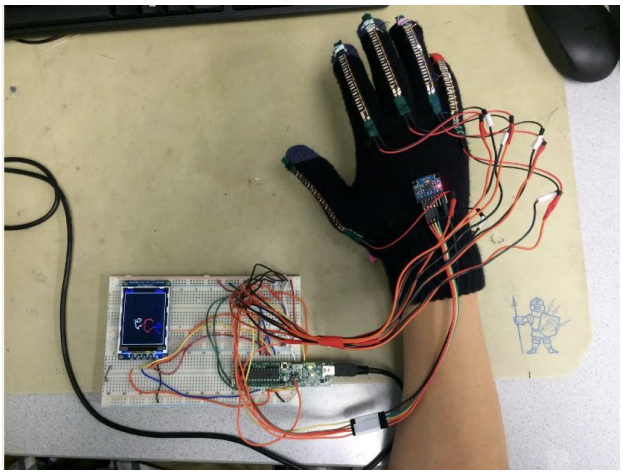
Fig. 11. WokWi Simulation of Speeh Unit.

B. *Recognition Accuracy and Latency*

- Gesture Recognition Accuracy: The embedded AI/ML classifier achieved gesture recognition rates of approximately 90–94% for the core set of trained hand signs under controlled conditions. Variability increased slightly with new users and under non-ideal lighting or background noise.
- Speech-to-Morse Pipeline: The speech-to-text algorithm maintained an 85–90% word recognition rate for clear speech; minor reductions were observed with accented or rapid speech.
- Latency: End-to-end response times (from gesture to speech output, or from spoken word to haptic motor vibration) ranged from 350–500ms, providing near real-time feedback suitable for interactive communication.

C. *User Study*

- The glove interface was comfortable and intuitive after short training.
- Most users were able to communicate basic phrases and receive tactile Morse feedback without difficulty.
- Feedback highlighted the system's portability and offline capabilities as valuable features.



D. *Limitations and Observed Challenges*

- Gesture Variability: Recognition dropped with highly personalized signing styles or rapid, continuous motions.
- Environmental Factors: Accuracy for speech recognition decreased in noisy conditions; sensor readings were sometimes affected by sweat or loose fitting.
- Hardware Constraints: Power consumption and battery life are areas for improvement, especially in long-use scenarios.
- Limited Vocabulary: Current prototype supports a limited set of gestures and spoken phrases; expanding the dictionary and supporting dynamic signs is a future goal.
- Haptic Feedback Learning Curve: Some users required time to interpret Morse code vibrations reliably, indicating a need for training or more intuitive tactile encoding.

VII. Conclusion

This work demonstrates the feasibility and effectiveness of Vanitra, a wearable, bidirectional communication system for deaf-mute users. By combining flex sensors, hall effect sensors, magnets, IMU, and the ESP32-S3 microcontroller with embedded AI, the glove unit reliably translates hand gestures into spoken words. The secondary module further enables communication by converting speech into Morse code haptic feedback, fostering two-way interaction without reliance on internet connectivity.

Prototype testing revealed high accuracy for gesture recognition and acceptable latency, with positive feedback from initial user studies. While limitations exist—such as vocabulary size, variability in signing styles, and environmental effects—the system proves itself a practical step toward inclusive, real-time communication.:

A. *Future work*

It will include several enhancements. Adding Force Sensitive Resistors (FSRs) will allow for more nuanced hand pressure sensing, potentially improving gesture accuracy and enabling recognition of a broader range of signs. Development of a companion module in the form of a smartwatch will offer users greater flexibility over language and sign language selection. This interface will let users easily choose which sign language system (such as ISL, ASL, or BSL) and spoken language they wish to use for communication, further customizing and broadening the device's applicability.

Additionally, ongoing improvements will aim to expand the gesture set, introduce more advanced speech synthesis, optimize hardware for power and comfort, and pursue extensive real-world testing to ensure effective everyday use.

B. Social implications are significant:

Vanitra has the potential to empower deaf-mute individuals by bridging communication gaps in various settings, from daily interactions to workplaces and public environments. Its offline function and affordable design promote broader adoption, leading toward greater inclusion, independence, and opportunity in deaf-mute communities.

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